Systemization of Knowledge (SoK): Visualization Insight – Two Decades of Research, Practice, and Future Directions

CHEN HE, Department of Computer Science, University of Helsinki, Helsinki, Finland
NIKLAS ELMQVIST, Department of Computer Science, Aarhus University, Aarhus, Denmark
ANDREA BELLUCCI, Department of Computer Science, Universidad Carlos III de Madrid, Leganés, Spain
MAHDI MUNSHI, University of Helsinki, Helsinki, Finland
GIULIO JACUCCI, Department of Computer Science, University of Helsinki, Helsinki, Finland

Insight generation has long been a primary goal of information visualization. Yet, there remains a lack of a comprehensive review on the research advancements of Visualization Insight. This systematic literature review analyzes nearly two decades of theoretical, empirical, and technological research development of visualization insight. By examining 340 articles from 17 leading journals and conferences in visualization, human–computer interaction, and databases and conducting an in-depth review of 92 papers, we propose a framework illustrating the iterative and collaborative process of generating, exploring, and communicating visualization insights. We then use the framework to analyze and compare the reviewed literature, revealing that (1) Research on visualization insight has evolved from defining insights to formalizing the insight generation process and communicating visualization insights through reports and data stories; (2) Issues like cognitive biases and chart errors hinder the effectiveness of visual discovery; (3) Automation in data fact mining and story generation eases data storytelling, but often lacks the incorporation of domain information, limiting the stories' impact. Moving forward, we outline five future research directions, aiming to improve insight quality and expand the methods of generating and communicating insights.

CCS Concepts: • **Human-centered computing** → **Visualization**; **Human computer interaction (HCI);**

Additional Key Words and Phrases: Information visualization, insight, empirical study, annotation, data story, data fact

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Authors' Contact Information: Chen He (corresponding author), Department of Computer Science, University of Helsinki, Helsinki, Finland; e-mail: chen.he@helsinki.fi; Niklas Elmqvist, Department of Computer Science, Aarhus University, Aarhus, Åbogade 34, Denmark; e-mail: elm@cs.au.dk; Andrea Bellucci, Department of Computer Science, Universidad Carlos III de Madrid, Leganés, Madrid, Spain; e-mail: abellucc@inf.uc3m.es; Mahdi Munshi, University of Helsinki, Helsinki, Finland; e-mail: mahdi.munshi@helsinki.fi; Giulio Jacucci, Department of Computer Science, University of Helsinki, Helsinki, Finland; e-mail: giulio.jacucci@helsinki.fi.



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1 Introduction

Since the 1980s, researchers have argued that the main goal of visualization is to help users gain insights from data [20]. For example, McCormick et al. [13] (1987) stated, "The goal of visualization is to leverage existing scientific methods by providing new scientific insight through visual methods." In the 90s, Ben Shneiderman adapted Richard Hamming's famous statement "The purpose of (scientific) computing is insight, not numbers", by highlighting that "The purpose of visualization is insight, not pictures," which guided researchers and practitioners to prioritize insight in visualization design. In 2005, Saraiya et al. [108] proposed an insight-based evaluation method to directly measure how well a visualization achieves the insight goal, distinct from the task-based evaluation, which measures task time and error. This study sparked increased discussion within the visualization community regarding defining insights [19, 96], developing insight-based evaluation techniques [51, 122], and sharing insights for collaboration and communication [120, 152]. With the rise of computational methods, data mining and machine learning techniques are utilized to extract insights from data, such as trends and outliers [40, 129], and automate data stories [117, 128].

However, challenges also arise from data-driven analysis. Users may derive spurious discoveries from charts without statistical evidence [150]; automated data insights often lack actionability [64]. With the popularity of Vision Language Models, their capabilities in image and text analysis are increasingly applied to tasks such as chart annotation [154] and data narrative generation [29]. The effectiveness of these applications hinges on a nuanced understanding of insight characteristics and appropriate evaluation methods. Notably, Karer et al. [64] caution against the data-centric view on analysis and prompt visualization systems to involve levels of context to help users attain **insights into the domain**. For instance, when examining a line chart showing annual CO_2 emission values, a user may notice that the highest and lowest values occurred in certain years. However, without relating to external information, the insight is just numbers, lacking informativeness. By linking these extremes to real-world events from those years, the user gains insight into the domain, which fosters learning and reflection.

Looking at the research on Visualization Insight over the years, we need to sort out **what new dimensions have emerged beyond the definition and categorization of insights (RQ1)**, including but not limited to how the insight-based evaluation has evolved and been utilized, what we understand about users' insight discovery process, and what algorithms and tools have been created to support visualization insight. Most importantly, we must consider **what are the future directions and key challenges in advancing the understanding and application of visualization insight (RQ2)**. To answer the **Research Questions** (**RQs**), we analyzed 92 papers on the theoretical (Section 4.1), empirical (Section 4.2), and technological (Section 4.3) development of visualization insight, screened from 340 articles retrieved across the fields of information visualization, **Human–Computer Interaction** (**HCI**), and database research (Section 3). From this analysis, we distilled a common process depicting the flow of visualization insight (Figure 1), which we used to compare and position the reviewed papers. We also examined how this process model relates to existing visual analysis and sensemaking models, highlighting its novelty and potential applications (Section 5). Finally, we identified five key research challenges aimed at enhancing the quality of visualization insight and the flexibility in the generation and transferability of visualization insight (Section 6).

The contributions of this **Systemization of Knowledge** (**SoK**) are three-fold: (1) We present an analysis of the state-of-the-art research on visualization insight from theoretical, empirical, and technological development viewpoints. (2) We propose a model to illuminate the iterative and collaborative process of generating, managing, and communicating visualization insights, flexible to accommodate diverse user roles and usage scenarios. (3) We shed light on future research directions to facilitate visualization insight. We believe this SoK will benefit multiple reader groups. Researchers can grasp the research development of visualization insight and identify promising

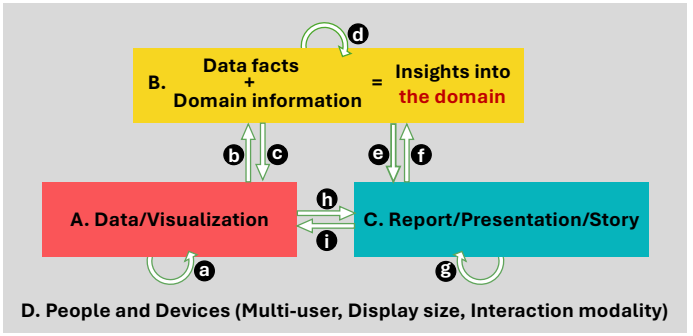


Fig. 1. A common process for visualization insight generation and communication, informed by 92 reviewed papers. It consists of four states: (A) data and their visualization, (B) an insight repository, and (C) a report/presentation/story, situated within the conditions of (D) people and devices, such as asynchronous collaboration and touch devices. Nine transitions connect the three states, indicating user intention shifts among the states. In different scenarios, the transitions embody different meanings. Figures 4–13 illustrate how the reviewed papers relate to this model. Here are examples of the transitions: (a) VDE; (b) insight externalization; (c) insight-driven VDE; (d) insight exploration, commenting, and recommendation; (e) report writing, data story creation, and dashboard generation; (f) generating insights from the story content; (g) report editing, styling, presentation, and reading; (h) automated report/story generation; (i) data story motivated VDE. Note that the process diagram is not a separation of visual space. For instance, users can (d) explore annotations in the visualization space (A).

directions for future work. Developers can understand what can be achieved and what can be further developed regarding the computational assistance for visualization insight. Students can learn the concept of visualization insight and the diverse topics developed in this area.

2 Background

This section discusses the terms used in this SoK and how this review differs from relevant reviews.

2.1 Definitions

Although visualization insight is often considered elusive, measuring time saved on an analytical task using visualizations is more straightforward [21]. However, time-based measures alone cannot capture important aspects of visualization, such as the open-ended nature of real-world data analysis. To evaluate insight enabled by visualization, a clear definition is required.

Researchers have therefore attempted to define visualization insight through literature reviews and empirical studies (Section 4.1.1). Chang et al. [19] and North [96] approached this task from multiple perspectives, proposing insight characteristics. Yang et al. [147] introduced the term "nuggets" to represent data queries and their corresponding results, proposing methods to identify important queries. Despite the varying definitions, Karer et al. [64] observed a unifying theme of visualization insight as **a step forward in the interpretation and analysis in the form of a change of the user's knowledge or understanding**. This is the definition we adopt in this review. According to the level of attainment, Karer et al. further distinguished visualization insight by insights into the visualization, data, and domain: **Insights into the visualization** are users' interpretations of the visualization, such as understanding visual encodings through legends; **insights into the data** (**data facts** as its synonym) reveal information about the data, such as correlations and trends; **insights into the domain** indicate insights that affect the viewer's understanding of the domain [64], as the example in Section 1 demonstrates. The goal of visualization

is to support insights into the data at least, as they come naturally after users obtain insights into the visualization. Accordingly, this review categorizes the focus of existing studies on whether they support insights into the data or the domain.

Visual Data Exploration (**VDE**), a fundamental method for supporting Exploratory Data Analysis [9, 132], enables users to interact with the data to uncover insights [149]. We adopt the definition of VDE as "the process of extracting insight from data via interaction with visual depictions of that data [60]" to unify the various terms used in literature, including Exploratory Visual Analysis [9] and Visualization Exploration [60]. While VDE enables users to think through visualization, Insight Externalization, as a follow-up to VDE, denotes "the process of capturing and recording the semantics of insights [25]." For instance, users mark a visual pattern on the chart, circling out the maximum point on a line chart, and narrate their discoveries orally or in writing. According to the reviewed papers, annotations and notes are two common forms of externalized insights: *annotations* are visual marks and texts embedded in charts, whereas *notes* are drawings and texts collected separately from the VDE space. Additionally, data story/storytelling links multiple insights in a logical and structured way to transform information into knowledge and convey a meaningful narrative [107].

2.2 Related Surveys

A recent survey by Battle and Ottley [10] proposed a formalism to represent insights and insight provenance in graph structures by reviewing 42 papers on defining and measuring visualization insight. However, beyond defining visualization insight, our review also covers the analysis of empirical studies and system & algorithm development on visualization insight, aiming to outline future research directions surrounding its theoretical, empirical, and technological development. This review also closely relates to surveys on visualization annotations [101, 102], which identify the design space of annotations and their utilities. The concept of visualization insight extends beyond annotations to encompass topics such as visualization evaluation, errors & biases in insight discovery, and insight management & recommendation.

Otherwise, reviews on data storytelling are relevant to our work. As an end product of data exploration, data stories closely relate to the communication phase of visualization insight (Figure 1(C)) [73]. Prior work has extensively reviewed data storytelling, including the design space characterization [12, 113], common elements in data stories [130], and automation to support storytelling [22, 79]. This review does not attempt to exhaust storytelling-related papers, as this has already been done by related surveys. We view data stories as an essential artifact in the cycle of visualization insight; our analysis results advise the involvement of domain information to enrich data stories rather than a simple logical connection of data facts.

3 Method and Analysis

We referred to the PRISMA 2020 statement [98] in reporting the process and results of this review.

3.1 Paper Selection

To define the scope of this review, we used a data-driven approach. We started from a set of 15 highly cited papers (Appendix A) that explicitly investigate visualization/data insights. By analyzing their publication venues and citation contexts, we identified three relevant research areas: Visualization, HCI, and Databases. Visualization insight gained increased research attention following the introduction of the insight-based evaluation method by Saraiya et al. (2005) [108]. To capture this development, we retrieved articles published between Jan 2005 and June 2024 from the leading journals and conferences in the three areas (Table 1), also including early-access articles,

Table 1. Data Sources of This Review

Area	Journal/Conference
Visualization	IEEE Transactions on Visualization and Computer Graphics (TVCG), IEEE Computer Graphics and Applications (CG&A), Computer Graphics Forum (CGF), Information Visualization – Sage journal (IV), IEEE Visualization (VIS), IEEE Visual Analytics in Science and Technology (VAST), Eurographics Conference on Visualization (EuroVis), IEEE Pacific Visualization (PacificVis)
HCI	ACM Transactions on Computer-Human Interaction (TOCHI), ACM Conference on Human Factors in Computing Systems (CHI), ACM Symposium on User Interface Software and Technology (UIST), ACM Conference on Intelligent User Interfaces (IUI), ACM Conference on Designing Interactive Systems (DIS)
Database	ACM Special Interest Group on Management of Data (SIGMOD), ACM Conference on Information and Knowledge Management (CIKM), Conference on Very Large Databases (VLDB), ACM Conference on Knowledge Discovery and Data Mining (KDD)

short papers, and extended abstracts. Our search was limited to top venues to ensure the inclusion of studies that underwent rigorous peer reviews and have relatively high impacts.

To retrieve diverse studies on visualization insight from these fields, we iteratively built a list of search queries by (1) extracting terms from titles of relevant articles that appeared in the search results and were retrieved via paper citation links, and (2) incorporating them into successive rounds of queries. The final set of terms used to search for document titles is *insight, annotation, knowledge, storytelling, report, fact, data story, note-taking, and comment.* As an additional condition, the term "visualization" must appear in either the full texts (for SIGMOD and VLDB only) or keywords (for HCI and the rest of the database-related sources) of the documents. We also performed backward and forward citation searches [140] to capture relevant work that fell outside of the search queries. The first author then screened the papers by assessing their titles, abstracts, and full texts (Figure 2) using the following inclusion and exclusion criteria.

Inclusion criteria. A paper must satisfy at least one of the following:

✓ The paper defines/characterizes visualization insight.
✓ The paper proposes a framework or method to capture insight discovery/knowledge generation.
✓ The paper (1) studies visualization on what insights people discover or how people discover insights, or (2) designs visualization to support various types of insights.
✓ The paper studies how people take notes during VDE or designs methods to support annotation/note-taking.
✓ The paper studies the correspondence between texts and charts or methods to offload analysis and thinking, such as linking notes and visualization.
✓ The paper addresses data fact mining or manual/automatic insight/report generation.

Exclusion criteria. A paper is excluded if it meets any of the following:

× The paper does not address the issue of discovering insight from the visualization in the design process, nor does it use an insight-based method to evaluate the visualization, such as designing a visualization to integrate knowledge [84] and automatic annotation of areas of interest [103].
× The paper is about storytelling but does not focus on the story content, such as addressing the design space [12] and character design in data stories [36].
× The paper focuses on text analysis rather than visualization insight, such as visualizing product reviews [146] and arguments [124].
× The paper is about domain knowledge input to support VDE rather than supporting insight discovery (e.g., [65]).

3.2 Paper Coding

Figure 3 shows the publication timeline of the reviewed papers. Out of the 92 selected papers, we have 63 papers from visualization-related venues, 21 from the HCI field, and 8 from database research.

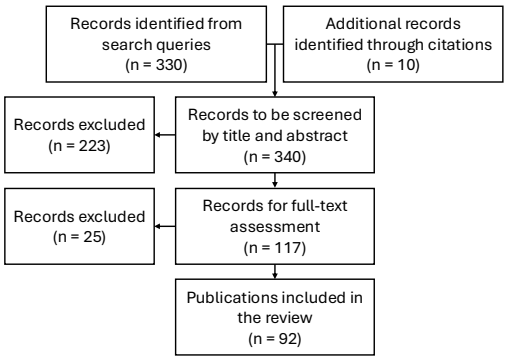


Fig. 2. Paper screening process.

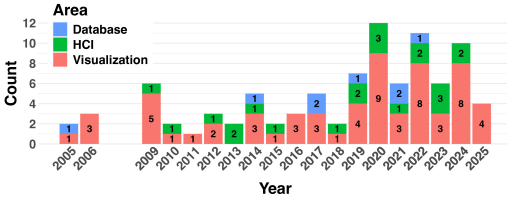


Fig. 3. Publication years of the 92 reviewed articles from the three areas of publication venues.

We coded the papers based on their insight levels (insight into the data/domain), contributions (theoretical, empirical, and technological research), and research topics. The coding criteria of insight levels and contributions are as follows.

Insight into the data. The paper focuses on data facts without mentioning the involvement of other contextual or domain information, such as devising methods/templates to extract/record data facts. *Insight into the domain.* The work highlights the role of contextual/domain information in visual discovery or allows users to externalize insights via note-taking and think-aloud, as in these cases, users can flexibly incorporate their domain knowledge.

Theoretical research. Papers that contribute to readers' theoretical understanding of visualization insight that could be generalized to visualization systems, such as defining visualization insight and modeling knowledge generation. *Empirical study.* Articles that focus on user studies and do not propose system designs, such as proposing insight-based evaluation methods. *Technological contribution.* This group of papers contributes *(1) visualization/database systems* to support insight discovery, externalization, & exploration and assist report generation, and *(2) algorithms*, such as data mining techniques to extract data facts and machine learning pipelines to generate stories.

The decision of the codes is data-driven, except for the insight levels, which result from a top-down approach referring to Karer et al. [64]. Two coders, authors of the SoK, collaborated on coding the papers. With an initial idea of coding the insight levels and contributions, one coder went through the papers, marking their insight levels and contributions and noting down the papers' purposes, methods, and findings. The coder then categorized the papers based on their topic similarities in each contribution group and provided a description for each topic category; the reading helped eliminate ambiguities in the categorization.

Following the coding criteria, two coders independently coded 15 randomly selected papers, accounting for the possibility that each paper could be relevant to multiple contribution groups and topics. While the coding of paper contributions showed complete agreement, topic coding was more challenging. We computed paper-level agreement using Jaccard similarity between coders' assigned topic sets. The mean Jaccard similarity across these papers was 0.46. Discussions helped refine the topic coding criteria. For instance, within the empirical study group, we decided to assign the paper to only one of the following topics: 1. examining insight as an evaluation method, 2. studying insight for tool design, and 3. how to support insight externalization, as we noticed that papers in topics 1 and 3 naturally suggest implications for tool design. Next, the two coders independently applied the refined criteria to another 25 randomly selected papers, resulting in an improved mean Jaccard similarity of 0.70. Another round of discussion was conducted to ensure

clarity and consistency in topic classification. Appendix B presents the final topic coding criteria. Finally, one coder coded the remaining papers using the finalized criteria.

3.3 Limitations

The limitations of this review include its scope and human subjectivity. We recognize that no systematic review can be exhaustive in capturing all relevant literature. To balance the quality and integrity of this review, we focused on 17 well-regarded sources from three research areas to ensure the resulting literature is of the highest significance, while still providing sufficient breadth for readers to grasp the state of research on visualization insight. As a result, some relevant studies may fall outside the scope of this review.

Using a data-driven approach to select and code papers can lead to variations in the collected literature, as different researchers may generate distinct query lists. Despite the efforts to standardize the review process, subjectivity in coding decisions may affect the accuracy of the review. To mitigate this, we employed two coders to ensure the consistency and reproducibility of the paper categorization.

4 Results

An interactive table overviewing the coding results, along with a **multidimensional scaling** (**MDS**) projection of the coded literature, is provided in Appendix C. Figure 1 presents a process model that summarizes the research on visualization insight, with more explanations in Section 5.

In this section, we present an analysis of the theoretical, empirical, and technological research on visualization insights by topic, using the summary model to illustrate the scope and focus of the reviewed literature. Notably, the coding of topics focuses on the methods and goals of the reviewed papers, whereas the proposed framework emphasizes the processes the research supports. As a result, semantically similar processes can serve distinct research goals.

4.1 Theoretical Research

Appendix D overviews the coding results of theoretical and empirical research papers. Eleven theoretical papers aim to characterize visualization insight or formalize the insight generation process.

4.1.1 Defining/Categorizing Insights. This group of work corresponds to Figure 1(B): **Defining insight** to support visualization design and evaluation, insight automation, and so on. Chang et al. [19] discussed two types of insights integrating the concept from cognitive science and the visualization field. Spontaneous insight (an "aha" moment) indicates a sudden change from a state of unknown to a knowing state; knowledge-building insight broadly refers to an advance in knowledge or a piece of information [108]. The two types of insights are interrelated as spontaneous insights occur with the accumulation of knowledge. Visualization studies must understand how both types of insights are promoted in solving real-world problems [19].

North [96] characterized insights as complex, deep, qualitative, unexpected, and relevant, whereas an important insight would rank high in all these characteristics. With a different purpose of automating insights, Law et al. [71] categorized 12 types of data facts resulting from four analysis tasks, including data wrangling, focused or exploratory analysis, and communication, through reviewing 20 insight automation systems. Similarly, Chen et al. [27] classified 12 types of facts for multi-dimensional data and summarized essential attributes for each insight type, such as a cluster's maximum and minimum values. This way, insights could be linked to data to benefit exploration, such as insight retrieval based on attributes of interest.

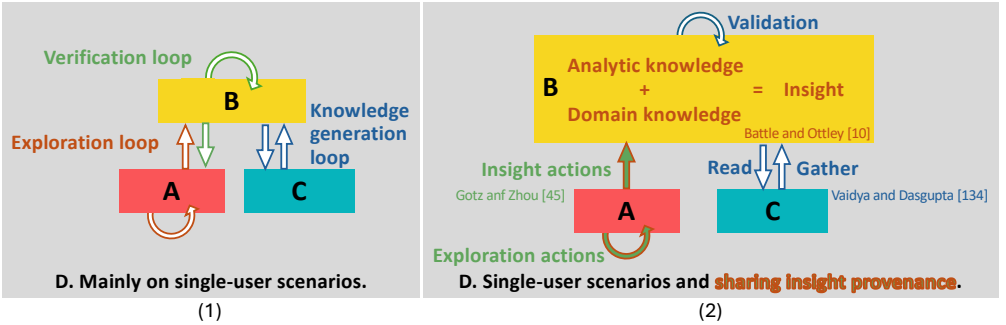


Fig. 4. How the reviewed articles in Section 4.1.2 relate to the framework. (1) The framework is used to represent the knowledge generation model [107] with the three interconnected loops. (2) Blue arrows indicate the reasoning framework by Vaidya and Dasgupta [134], where users read communicative charts and gather evidence to validate facts; Green arrows map to Gotz and Zhou's taxonomy of exploration and insight actions to capture insight provenance [45]; Orange part depicts Battle and Ottley's formalism of visualization insight as a collection of analytic knowledge (information gathered from visual analysis) and domain knowledge [10]. Sharing insight and provenance data can facilitate collaboration and analysis reuse [10, 45].

However, Karer et al. [64] questioned the practice of data-centric analysis and necessitated the involvement of the levels of context to obtain **insights into the domain**. Without contexts, such as users' analytic tasks and domain knowledge, insights into the data are not applicable; integrating domain knowledge in visualization is essential to validate the findings against the domain. Similarly, Law et al. [72] identified seven insight characteristics by interviewing 23 professional users of visualization platforms. More than a piece of information, insights are often paired with domain knowledge or other contextual information and pieced together based on users' intuition [72].

From a cognitive science viewpoint, Smuc [121] proposes three levels of insight, bridging the gap between lower-level perception and higher-level sensemaking tasks. With increased levels of cognitive processing, skill-level insights result from internalized skills, rule-level insights are based on rule sets, and knowledge-level insights come from reasoning, abstracting, or knowledge transfer. This clearly shows that visual discovery is closely tied to contexts, including user skills, experiences, and prior knowledge.

4.1.2 Formalizing the Insight Generation Process. Apart from studying the resulting insights, researchers also endeavored to understand the insight generation process. Sacha et al. [107] proposed a knowledge generation model to guide visualization design and evaluation. Mapping to the cycles of visualization insight (Figure 4(1)), the model includes **the exploration loop** to discover data facts, **the validation loop** to interpret insights in the context of the domain, and **the knowledge generation loop** to collect enough insights to be accepted as new knowledge. As Sacha et al. pointed out, the validation and knowledge generation loops closely relate to the creation of reports and references to visualizations. Another framework by Vaidya and Dasgupta [134] addresses the decoding process of communicative visualizations in which users collect insights from published charts to associate back to existing facts to support information validation (blue arrows in Figure 4(2)). To facilitate chart interpretability, they suggest annotating charts with visual cues & contextual explanations and avoiding missing information.

Besides examining **human cognition** to understand insight generation, researchers also suggested capturing **the actual analysis process** by logging user interactions [45] and data transformations [10]. As Xu et al. [145] observed, analysis provenance closely relates to the generation of insights and reports. To capture this process, researchers have represented provenance data using

graph structures [10, 45]. Based on a four-tier characterization of user interactions, Gotz and Zhou [45] used the action tier, general across the domain while semantically rich, as building blocks to record insight provenance and proposed an action taxonomy on exploration, insight, and meta actions (green arrows in Figure 4(2)).

However, prior studies hinted that tracking data transformations can be advantageous over interaction logs [10]. Battle and Ottley [10] record user domain knowledge and knowledge gained from data analysis (i.e., data relationships derived from data transformations) to represent knowledge expansion in graphs (orange part of Figure 4(2)), which could be applied to measure knowledge depth and breadth, compare users' insight generation, and reuse the analysis to extract knowledge from new datasets.

4.1.3 Summary and Discussion. A visualization insight, be it spontaneous or based on knowledge building, contains multifaceted characteristics, such as being qualitative and unexpected. Categorizing *insights into the data* can benefit insight automation and management, but without the involvement of domain knowledge, insights lack actionability. This raises the challenge of **how to effectively incorporate contextual information in visual analysis** to support actionable insights and the generation of knowledge [10, 72, 107]. Further, researchers have characterized the insight generation process from human cognition, interaction, and data transformation viewpoints, primarily focusing on single-user settings. These frameworks hold potential for applications in visualization evaluation, inferring higher-level tasks, and analysis reuse. Nonetheless, expanding the focus to **collaborative analysis scenarios** could further evaluate and extend these frameworks.

4.2 Empirical Studies

We categorized empirical studies into five topics (Appendix D):

4.2.1 Examining Insight as an Evaluation Method. Consistent with the development of theoretical papers, insight-based evaluation methods have grown throughout the years, first assessing the insights alone and then analyzing the insight generation process. Saraiya et al. [108] proposed the insight-based evaluation that studies what insights the visualization enables users to discover by analyzing the characteristics of user-generated insights, such as insight categories and domain values, distinct from the traditional task-based evaluation, which measures task time and error to compare visualization designs. With open-ended exploration, insights can be collected through the think-aloud protocol or note-taking and then coded across multiple aspects to compare the tools' capabilities (Figure 5). Saraiya et al. further conducted a longitudinal study to capture insight generation in a realistic environment, gaining knowledge on how to improve the tools [109].

North [96] critically evaluated the use of benchmark tasks to assess visualization insight; predefined tasks could induce bias and limit users' creative thinking. This led to the conclusion that an open-ended protocol, removing benchmark tasks, facilitates qualitative insight analysis, such as assessing the depth and unexpectedness of insights. On the other hand, Gomez et al. [44] proposed a combined evaluation with predefined tasks before open-ended exploration to evaluate both the task performance and insight generation capabilities of visualization. They choose low-level tasks that would not steer participants toward insights.

However, by comparing the task and insight evaluation methods, North et al. [97] found that often, a significant link exists between the task performance and the number of that type of insight users produce; the insight method also contains additional, higher-level insight categories that were not considered by the tasks. This indicates that predefined tasks may not well represent what will occur in real-life tasks; to know what users would do with the visualization, the insight method is a better measure. Apart from studying individual insights, Smuc et al. [122] analyzed how an insight was built upon prior knowledge/other insights on a timeline, which helped to understand

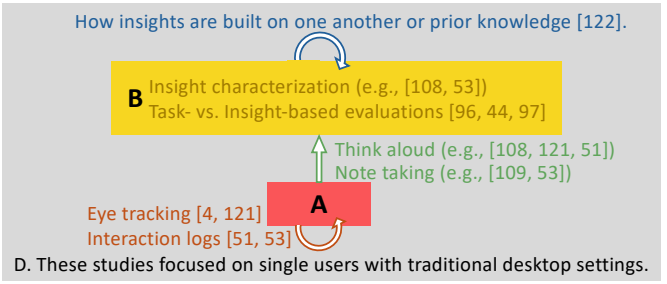


Fig. 5. Papers that examine insights as visualization evaluation methods (Section 4.2.1). They collect eye tracking [4, 121] or interaction logs [51, 53] to inform the insight generation process, and use think-aloud protocols (e.g., [51, 108, 121]) or note taking (e.g., [53, 109]) to record insights. Insights are characterized on multiple aspects such as domain values and correctness (e.g., [53, 108]). Insight measures are compared against [96, 97] or combined [44] with task-based methods to evaluate visualizations, or assessed based on how they build upon prior information [122].

where a user succeeds or fails in gaining the intended insights and was found useful for early-stage prototype evaluation in participatory design.

Other studies investigated the insight generation process by assessing eye tracking and interaction logs besides externalized insights (Figure 5). Upon studying how participants generate insights during VDE using eye-tracking and think-aloud data, Andrienko et al. [4] discovered that prior knowledge helps visual pattern interpretation, and eye-tracking data can reveal tacit analysis that is not verbally narrated. For example, gaze patterns indicate different analysis strategies, such as hypothesis-driven and compilation. Smuc [121] also employed eye tracking, along with the think-aloud protocol, during users' open-ended VDE sessions to understand their cognitive processing and insight generation. Results show that pupil diameters can reveal high or low levels of cognitive processing, but cannot clearly distinguish the proposed three levels of processing: skill-, rule-, and knowledge-based.

Two studies investigated the correlations between interaction types/patterns with insights. By correlating interactions to three types of insights (facts, generalizations, and hypotheses), Guo et al. [51] identified the interactions that promote/inhibit insights. He et al. [53] correlated interactions with insight directedness versus unexpectedness, breadth versus depth, and domain values. The results imply the application of automatic insight characterization and insight recommendation through user interactions.

4.2.2 Studying Insight for Tool Design. Besides examining insight as an evaluation method, researchers studied how people use existing tools to generate insights to inform visualization design, highlighting the practical significance of insight-based evaluation methods. Kandogan et al. [63] interviewed business analysts on how they discover and deliver insights, which relates to the complete cycles of visualization insight and collaborative analysis in practice (Figure 1). Results reveal that data analysis in practice is complex and segmented, involving heterogeneous data sources and people with different expertise using various tools; support of the fluid transition between analysis and presentation and advanced metadata management, including analytic provenance management, is required.

To determine the design requirements of tools supporting non-data experts to draw and communicate insights from their personal data, Choe et al. [31] analyzed 30 video recordings on the presentation of self-tracking data (Figure 6(1)). Results suggest future tools to (1) encourage self-reflection on their data, (2) incorporate statistical analysis to help validate insights, (3) suggest

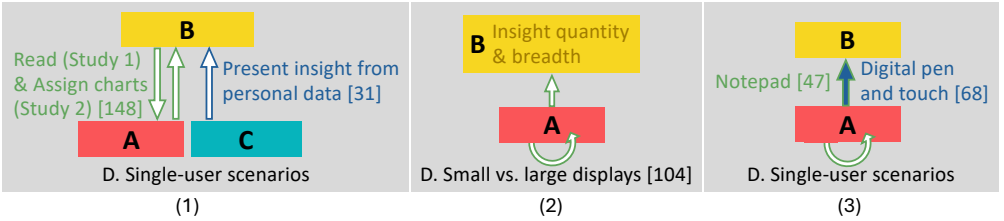


Fig. 6. Papers that examine visualization insight to inform tool design (1-2) and study insight externalization (3). (1) Yang et al. [148] investigated the types of insights users derive from reading charts and which charts users assign to derive certain types of insights to inform the design of visualization recommenders; and Choe et al. [31] studied how users present insights from personal data using various charts to inform the design of tools that support non-data experts to draw and communicate insights. (2) Reda et al. [104] studied how display sizes affect insight discovery. (3) Goyal et al. [47] studied the use of interactive visualizations and a notepad to support analysis tasks, while Kim et al. [68] examined the use of digital pen and touch interactions to link charts and notes for insight externalization.

the best charts for an insight, and (4) support interactivity, animation, and visual annotation for presentation. Meanwhile, to recommend charts for deriving a particular type of insight, Yang et al. [148] conducted two crowdsourced studies of bar and line charts and their compositions to understand what insights users derive from the charts (Study 1) and how users rank the charts for deriving specific insights (Study 2).

Additionally, by studying how display sizes (small and large) affect the quantity and breadth of insights (Figure 6(2)), Reda et al. [104] found that larger displays can significantly increase the number of discoveries while yielding broader, more integrative insights, as users are less likely to modify the visual layout, which helps sustain their focus on insight generation.

4.2.3 How to Support Insight Externalization. Researchers examined insight externalization in singular and collaborative settings with digital pen & paper and traditional note-taking methods (e.g., Figure 6(3)). To understand how to support people taking notes during VDE, Kim et al. [68] explored the use of a digital pen for insight externalization and touch interactions to link notes and visualization. Compared with conventional paper & pen or word processors, users generated more graphical annotations and more notes linked to visualizations. Similarly, upon studying the utility of graph visualization and a simple notepad for collecting notes in solving an analysis task, Goyal et al. [47] revealed that the visualization significantly improved participants' performance, whereas the notepad did not. Referring to Kim et al.'s discovery [68], the use of a digital pen with a linking feature may enhance note-taking performance.

On the other hand, Mahyar et al. [89] studied note-taking in a collaborative scene with tabletop and wall displays, which involved all three states in the framework (Figure 1). Results suggested the need for built-in note-keeping support, including linking notes to visual elements and note management features, as well as the necessity of both private and public notes to support collaboration. Learning from these studies, linking texts and visual elements appears to be a critical feature in supporting insight externalization.

4.2.4 Effects of Linking Texts and Charts. Of the work we collected, studies of linking texts and charts relate to understanding data stories and informing visualization design methods (Figure 7(1)). By comparing the performance of static visualizations and the corresponding data stories (static visualizations with visual highlights and textual annotations) in **data retrieval** and **insight comprehension** tasks, Shao et al. [115] found that data stories can enhance task accuracy regardless of participants' visualization literacy but may not necessarily improve task time.

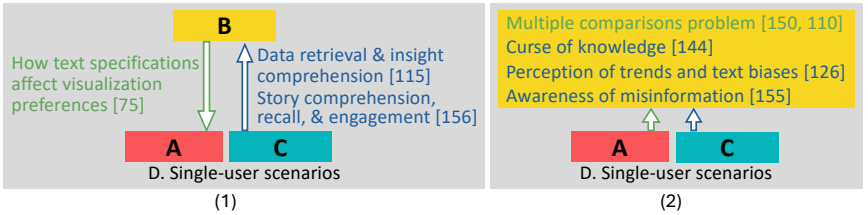


Fig. 7. Papers that study the effects of linking texts and charts (1) and challenges that undermine visual discovery (2). (1) Prior work studied how linking texts and charts affects data retrieval and insight comprehension tasks [115], as well as story comprehension, recall, and engagement [156]; Lee-Robbins et al. [75] investigated how different types of text specifications including learning objectives, insights, and assessment questions influence visualization design choices. (2) Studies also examined the multiple comparisons problem in VDE [110, 150], the impact of prior knowledge on the perception of visual patterns [144], and the relationship between texts and charts in users' perception of the information conveyed [126, 155].

Also relating to data stories, Zhi et al. [156] compared the spatial layouts (vertical and horizontal) and linking effects (linking and no-linking) of textual narratives and the visualization on user **comprehension** of the data story, **recall** of the story information, and **engagement** with the story. Results show that linking the textual content with the visual elements through mouseovers improved perceived engagement; for comprehension, the horizontal layout is more effective than the vertical one; for recall, with linking, the horizontal layout is better than the vertical one. Combining with earlier mentions, we can say that linking texts and visual elements could benefit both the authoring and reading of insights.

Moreover, through exploring the effect of three specification types, learning objective, insight, and assessment, on user preferences in visualization designs, Lee-Robbins et al. [75] found that specifying the learning objective can best support designers in choosing a more effective design.

4.2.5 Insight Discovery Fallacies. Many factors can affect the effectiveness of visualization insight discovery, including the multiple comparisons problem, cognitive biases, and chart errors (Figure 7(2)). By investigating the multiple comparisons problem in VDE, Zgraggen et al. [150] discovered that over 60% of user discoveries were false. Statistical correction, confirming with a different dataset or testing statistics on both implicit (unreported) and explicit (reported) insights, can lower the false discovery rate to around the bound of 5%. Further, Sarma et al. [110] found that uncertainty representations can improve users' decision quality and control false discovery rates.

Three articles explored how visual stories linking texts and charts affect readers' information uptake (blue part of Figure 7(2)). By studying the effect of prior knowledge on the perception of pattern saliency, Xiong et al. [144] found that when primed to see visual patterns as salient via visual cues or stories, people tend to think these patterns are also evident to uninformed viewers, which explains why sometimes experts fail to communicate visual patterns to the audience. Stokes et al. [126] revealed that textual annotation does not influence user prediction on the trends shown in the charts, but whether the texts align with user prediction can affect user appraisal of their degree of bias: When the texts do not match user prediction, users tend to rate texts as more biased.

Zheng and Ma [155] compared the effects of text-chart integration methods with (1) annotation on top of charts and (2) interactive linking between spatially separated text and charts on reader awareness of three types of misinformation (contradictory trends between charts and texts, truncated axis, and inverted axis). They found that **the majority remained unaware of misinformation**, but the two techniques could promote user awareness of misinformation and significantly lowered the perceived credibility of texts and charts compared with the static story

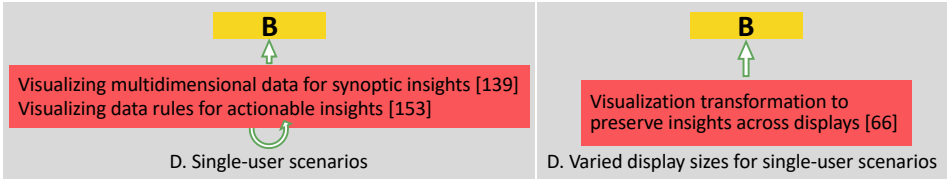


Fig. 8. Papers that focus on visualization design to support insights [139, 153] and visualization transformation to preserve insights across displays [66].

without explicit links between texts and visualization. Combining the results of the previous two studies, we speculate that users may overlook misinformation in charts even when the textual annotation conveys the truth. Fortunately, Section 4.3.6 collected some visual ways to effectively combat chart-induced misinformation.

4.2.6 Summary and Discussion. Insight-based evaluation methods, from characterizing visualization insight derived from VDE to analyzing the insight generation process using interaction and eye-tracking data, are fundamental in visualization evaluation and understanding users if the purpose of the visualization is to support insight. However, these methods mainly focus on single-user desktop settings; extension to collaborative and everyday data analysis scenarios is required, considering data analysis in practice is collaborative and involves various tools [63].

To inform tool design, such as supporting analysts to discover and deliver insights and devising visualization recommenders for depicting various types of insights, researchers (1) interviewed analysts on their data analysis practices and (2) studied what insights and how users derive insights from visualization. Their discoveries could enhance future tool development.

Researchers (1) explored insight externalization using digital pens, external editing methods, and touch interactions in singular and collaborative settings and (2) studied the benefits of relating texts and charts through, e.g., interactive linking and textual annotation in supporting user engagement with the information and awareness of misinformation. Results reveal, among others, the importance of linking texts with visual elements to support both the authoring and reading of insights. However, beyond note-taking, other outputs and interaction modalities, such as physiological signals, could be investigated to broaden the scope of insight externalization. Alternative environments, such as extended realities, could also be studied on how users externalize insights and make sense of data stories, potentially within the context afforded by the environment.

Moreover, many potential threats exist undermining effective insight discovery, including the multiple comparisons problem and cognition-induced & texts/charts-induced biases. Human cognition plays a critical role in insight discovery but is largely unexplored. Future research could examine various types of cognitive biases in visual discovery and how to mitigate their effects [39].

4.3 Technological Research

Table 2 illustrates the codings of technological research papers by insight level, contribution, and research topic.

4.3.1 Visual Representation to Support Insight Generation. Section 4.2.2 discussed empirical studies to advise tool design in supporting insight. In contrast, this section presents how researchers devised systems to support insight (Figure 8). PolyCube [139] employs multiple techniques, including superimposed views, juxtaposed views, 3D cubes, and animation (slideshow) to link multidimensional data, aiming to support an integrated understanding. A qualitative study reveals the

Table 2. Coding of Technological Research Papers

	Insight level		Contribution		Topic								
Articles	Insight into the data	Insight into the domain	Visualization design/ Database system	Algorithm	Visual representation to support insight generation	Mechanisms to support insight externalization	Insight management, exploration, and recommendation	(Automatic) Integration of texts and charts	Linking insights, tags, visualization/data, etc.	Supporting reflections on visualization/insight	Report/Presentation generation	Mining data facts for visual recommendation/storytelling	Machine learning for data fact/story generation
PolyCube [139] (CG&A20)		✓	✓		✓								
Opportunity Map [153] (CIKM05)		✓	✓		✓								
Kim et al. [66] (TVCG22)	✓			✓	✓								
ChartAccent [105] (PacificVis17)	✓		✓			✓							
ActiveInk [106] (CHI19)		✓	✓			✓							
Elias and Bezerianos [41] (CHI12)		✓	✓			✓	✓						
Click2Annotate [25] (VAST10)	✓		✓			✓	✓						
Touch2Annotate [26] (CHI EA10)	✓		✓			✓	✓						
FacetNotes [7] (IV22)		✓	✓				✓						
Shrinivasan and Gotz [119] (CHI EA09)	–[a]			✓			✓						
InsightNotes [143] (SIGMOD14)		✓	✓				✓						
Lai et al. [69] (CHI20)	✓			✓				✓					
Kori [70] (TVCG22)		✓	✓	✓				✓					
Cai et al. [15] (TVCG25)	✓		✓	✓				✓					
VizFlow [127] (CHI21)		✓	✓					✓			✓		
Charagraph [91] (CHI23)	✓		✓					✓					
Gotz et al. [46] (VAST06)		✓	✓			✓	✓		✓				
Dadzie et al. [35] (IV09)		✓	✓			✓	✓		✓				
Annotation Graphs [151] (TVCG17)		✓	✓	✓		✓	✓		✓				
Zhao et al. [152] (TVCG18)		✓	✓			✓	✓		✓				
CoVA [81] (CGF20)		✓	✓			✓	✓		✓				
InsideInsights [92] (CGF19)		✓	✓			✓	✓		✓		✓		
Fan et al. [43] (CHI22)	***[b]		✓							✓			
VisuaLint [56] (CGF20)			✓							✓			
McNutt et al. [94] (CHI EA20)		✓	✓							✓			
Maldonado et al. [90] (CHI20)		✓	✓					✓			✓		
Temporal Summary Images [14] (TVCG17)	✓		✓	✓				✓			✓		
Contextifier [58] (CHI13)		✓		✓				✓			✓		
Roslingifier [118] (TVCG23)	✓		✓								✓		
DataParticles [16] (CHI23)		✓	✓					✓			✓		
CrossData [28] (CHI22)		✓	✓					✓			✓		
VisInReport [114] (TVCG22)		✓	✓								✓		
GeoExplainer [76] (TVCG24)		✓	✓	✓							✓		
StoryFacets [99] (IV22)		✓	✓								✓		
Gratzl et al. [48] (CGF16)		✓	✓								✓		
Chen et al. [24] (TVCG20)		✓	✓								✓		
Tang et al. [129] (SIGMOD17)	✓			✓								✓	
QuickInsights [40] (SIGMOD19)	✓			✓								✓	
MetaInsight [86] (SIGMOD21)	✓			✓								✓	
Voder [125] (TVCG19)	✓		✓				✓	✓				✓	
Kandogan [62] (VAST12)	✓		✓	✓			✓					✓	
DataSite [34] (IV19)	✓		✓	✓			✓					✓	
LEVA [154] (TVCG25)		✓	✓	✓			✓				✓	✓	
Foresight [37] (VLDB17)	✓		✓	✓			✓					✓	
Boomerang [74] (SIGMOD21)	✓		✓	✓			✓					✓	
CoInsight [77] (TVCG24)	✓		✓	✓			✓				✓	✓	
Notable [80] (CHI23)	✓		✓	✓							✓	✓	
InkSight [83] (TVCG24)	✓		✓	✓							✓	✓	
DataShot [136] (TVCG20)	✓		✓	✓							✓	✓	✓

(Continued)

Table 2. Continued

Articles	Insight level		Contribution		Topic								
	Insight into the data	Insight into the domain	Visualization design/ Database system	Algorithm	Visual representation to support insight generation	Mechanisms to support insight externalization	Insight management, exploration, and recommendation	(Automatic) Integration of texts and charts	Linking insights, tags, visualization/data, etc.	Supporting reflections on visualization/insight	Report/Presentation generation	Mining data facts for visual recommendation/storytelling	Machine learning for data fact/story generation
Calliope [117] (TVCG21)	✓		✓	✓							✓	✓	✓
Calliope-Net [23] (TVCG24)	✓		✓	✓				✓			✓	✓	✓
Erato [128] (TVCG23)	✓		✓	✓							✓	✓	✓
Socrates [141] (TVCG24)	✓		✓	✓							✓	✓	✓
SNIL [29] (TVCG25)	✓		✓	✓							✓		✓
Sintel [1] (SIGMOD22)		✓	✓	✓								✓	✓
InsigHTable [78] (TVCG25)	✓		✓	✓			✓				✓	✓	✓
DashBot [38] (TVCG23)	✓		✓	✓			✓				✓	✓	✓
Shi et al. [116] (PacificVis24)	✓			✓							✓	✓	✓

[a]This work proposes an algorithm that recommends insight based on the interaction trails that lead to the insight, without mentioning the content of the insight.
[b]These two studies focus on insight into the visualization, annotating chart errors.

benefits of 3D views over coordinated views in supporting insight, contrary to the general criticism within the visualization community regarding 3D visualization.

Upon mining rules from data (i.e., the effects of data attributes on classes), the Opportunity Map [153] shows the relation between data attributes and classes in a table with rules visualized in table cells indicating their strengths; four regions of the table separate the actionability of visualization insight with user-defined important/less important classes and actionable/non-actionable attributes.

Kim et al. [66] proposed measures to preserve three types of insights (identification, comparison, and trend) when transforming visualization from desktop to smartphone displays. Training on expert-ranked small-screen visualizations for a variety of source views, their approach achieved 84% accuracy in ranking visualizations.

4.3.2 Mechanisms to Support Insight Externalization. Researchers created techniques and templates for insight externalization with varied interaction modalities (1. orange part of Figure 9). Ren et al. [105] created a design space of chart annotations by reviewing 106 annotated charts published by news graphics desks, identifying annotation forms and targets; ChartAccent, a tool resulting from the design space, supports manual, data-driven annotations with texts, shapes, highlights, and images. ActiveInk [106] utilizes pen and touch interactions to support a seamless transition between VDE and insight externalization. A qualitative study comparing the techniques of assigning pen actions before and after inking, such as assigning actions of highlighting and labeling before and after inking, indicates that prefix affords rapid sequential actions, such as drawing multiple highlights, whereas postfix better integrates with annotation and note-taking.

Research also addressed **data-aware insights**, where insights are linked to the data, facilitating insight retrieval from different views of the same data [54]. Elias and Bezerianos [41] focused on context-aware annotations, in which analysts can create and share annotations linked to multiple data targets, set annotation lifetime and validity, and retrieve annotations based on the linked data attributes. Similarly, upon user selection of data from charts, Click2Annotate [25] and Touch2Annotate [26] help users semi-automatically annotate data through templates of various insight types, with data attributes and narrative sentences readily available. On the other hand,

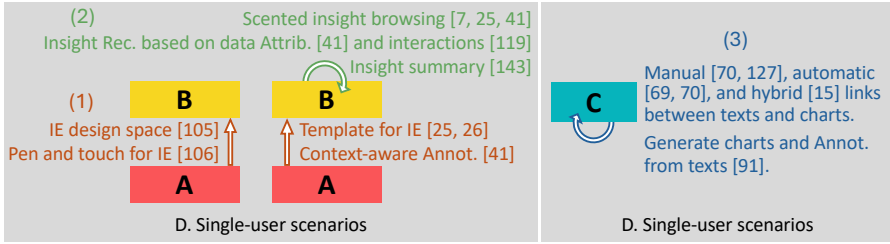


Fig. 9. Papers proposing methods to support insight externalization (1. orange part), exploration, and recommendation (2. green part). Prior research introduces design spaces for chart annotation [105] and methods for insight externalization (IE) using pen-and-touch interactions [106] and templates [25, 26]. Data-aware insights enable insight exploration from different views of the same data [7, 25, 41]. Research also addresses insight recommendation using interactions [119] and insight management with dynamic summaries [143]. To enhance reports (3. blue part), researchers proposed methods to generate charts from texts [91], as well as manual [70, 127], automatic [69, 70], and hybrid [15] ways to link texts and charts.

utilizing the links between data and insights, tools are created to construct insight graphs for sensemaking and collaboration [35, 81, 92, 151, 152], which we will discuss later.

4.3.3 Insight Management, Exploration, and Recommendation. Mainly studied in their textual form, notes and annotations contain metadata, such as the author, timestamps, and associated data attributes, facilitating insight management, exploration, and recommendation (2. green part of Figure 9). As mentioned earlier, data-aware insights link the narratives to data attributes, making insights easily accessible from charts (e.g., [7, 25, 41]). As an example, FacetNotes [7] shows lists of annotations over a chart, which are grouped and ordered based on their related data attributes. The lists adapt to user interactions to assist data exploration. For instance, annotations over a histogram are grouped by the bars and ordered based on the data coverage of the annotations; filtering of the histogram updates the annotations to display only those relevant to the filtered data attributes.

Further, insights could be recommended based on the similarity measure of data attributes [41] and interactions [119]. Shrinivasan and Gotz [119] utilize user interactions to link past notes to the current analysis. Each note has an action trail attached, recording data characteristics explored in both back and forward action traces to assess the relevance to the current line of inquiry. Additionally, focusing on database management, InsightNotes [143] manages large quantities of raw annotations through annotation summaries and provides a query engine to manipulate and propagate the summaries. Summaries can be text snippets, representative contents from clustering, and user-specified classes with the numbers of associated annotations attached. Further, Section 4.3.8 presents several systems that mine facts from data and recommend them to users with matching charts and visual highlights.

4.3.4 (Automatic) Integration of Texts and Charts. Studies in Sections 4.2.3 and 4.2.4 revealed the importance of linking texts and visualizations in supporting authoring and reading visualization insight. However, unlike data-aware insights, these links are often not explicit. To tackle this, this group of research features automatic and interactive methods to integrate texts and charts (3. blue part of Figure 9).

Taking charts as images, Lai et al. [69] place texts within charts with the corresponding visual highlights. Through visual entity detection and natural language processing, they achieved 59% annotation accuracy. Meanwhile, Kori [70] designs manual and automatic ways to link chart elements with text snippets, which include point- and interval-level references. It extracts chart

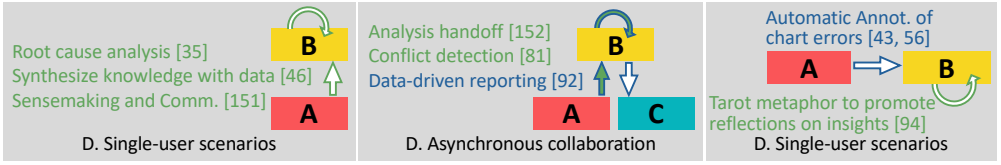


Fig. 10. Papers that link insights, tags, charts/data, etc. in graph structures to support sensemaking and collaboration (1-2) and that support reflection on charts and insights (3). Prior research links insights, tags, and charts/data to serve a range of analytical goals, such as root cause analysis [35], analysis handoff [152], analysis conflict detection [81], and data-driven reporting [92]. Additionally, researchers proposed automated annotations to highlight chart errors [43, 56], as well as the use of the metaphor of Tarot cards to promote reflection on automated insights [94].

features from its Vega-Lite specifications to match text snippets. An algorithmic evaluation indicated low accuracy (41.6%), whereas a user study implied the usability and usefulness of the tool.

Unlike Lai et al. and Kori, Cai et al. [15] utilize the collaboration between humans and machines to link texts and charts. They represent visualization as **knowledge graphs** (**KG**) linking data and visual elements, extract information from texts, and bind the information to KG entities. Users can edit the results of automatic linking. Compared to Kori's method, the reference detection algorithm has improved in accuracy to 59%. Another study enables users to manually link text snippets and chart highlights to animate data stories [127]. By choosing a dynamic layout, such as text snippets over or next to the chart, users can view the article flow via scrolling.

In contrast, to help readers make sense of data in texts, Charagraph [91] enables users to generate interactive charts (bar, line, and pie charts) from selected textual data. For instance, upon selecting data from a table, users can create and customize charts, such as editing labels, calculating statistics, and combining multiple charts. Two studies confirmed its usability and utility: Compared with a normal text reader, Charagraph produced more correct answers and required less effort on data analysis questions. Additionally, researchers developed other techniques to integrate texts and charts, including rule-based methods to map insights to visual highlights [90, 125] and algorithms to place annotations within visualization in a visually pleasing manner [14, 23].

4.3.5 Linking Insights, Tags, Visualization/Data, etc. Researchers consistently used tree [92] and graph (e.g., [151, 152]) visualizations to link insights, tags, charts, and so on for sensemaking and collaboration (Figure 10(1)–(2)). They enable users to create insights, such as concepts and instances, which are manually [35, 81, 92, 152] or automatically [46] linked with existing insights and data to record the visual discovery process. For instance, Gotz et al. [46] link newly created concepts with existing data through multiple visual metaphors, such as timelines and maps.

The generated analysis graph can serve diverse purposes, including root cause analysis [35], analysis handoff in asynchronous collaboration [152], and conflict detection of analysts' graphs [81]. For instance, Zhao et al. [152] enable the interactive playback of the graph creation process to support inferring the rationale of prior collaborators' analysis. By storing the insight graph as JSON structures, CoVA [81] uses Git to merge results and detect conflicts from different analysts. With more automation, Annotation Graphs [151] organizes user comments, tags, and data selections in various graph layouts to support sensemaking and communication. Node pinning and merging functions enable users to align the graph to their mental models.

However, tree structures are shown to be less cognitively demanding than graphs for users with little knowledge about the content [92]. To integrate collaborative analysis and reporting, InsideInsights [92] allows users to iteratively construct a hierarchical pipeline to record analysis

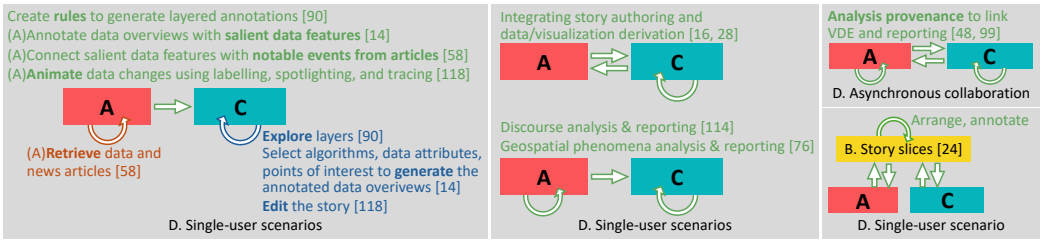


Fig. 11. Papers that build annotated time series visualization and animation for storytelling (1), support report authoring (2), and utilize analysis provenance or the insight space to mediate analysis and story creation (3). (A) indicates automation.

provenance and organize insights into hierarchies to support levels of detail. Each insight can be linked to the states in the pipeline and analysis views to facilitate presentation.

4.3.6 Supporting Reflections on Visualization/Insight. As discussed in Section 4.2.5, threats like the multiple comparisons problem and chart errors can undermine insight discovery. Researchers proposed approaches to assist users in identifying chart errors and reflecting on data patterns (Figure 10(3)).

Visual annotation is effective in helping users spot chart errors and avoid misinformation. Fan et al. [43] proposed a pipeline to annotate deceptive line charts with the correct one regarding y-axis truncation, inversion, and aspect ratio. Similarly, VisuaLint [56] sketchily rendered five chart construction errors to enhance chart comprehension, including size encoding with negative data, ineffective colors, missing legends, misaligned scales in dual-axis charts, and truncated axes. Crowdsourced studies verified their effectiveness, especially to laypeople [43, 56]. Scaffolding, providing textual explanations alongside visual ones, is necessary for lay users to understand unfamiliar and complex errors [56]. In contrast, McNutt et al. [94] employ the metaphor of Tarot cards to represent automated insights in a manner that encourages critique and reflection, with the Minor Arcana illustrating various data patterns and the Major Arcana posing questions to promote skepticism thereof.

4.3.7 Report/Presentation Generation. This section of work features manual-to-automatic report writing/generation and the smooth transition between VDE and reporting. Three articles place annotations over a base layer, typically a timeline to communicate data and insights (Figure 11(1)). However, the methods for the base layer and annotation generation differ. For educational purposes, Maldonado et al. [90] layer meaningful insights over a static background to provide explainability. In nursing simulation training, teachers can create rules, such as the position of nurses, as an annotation layer on a timeline of actual activities to facilitate student learning and reflection. With more automation, Temporal Summary Images [14] summarizes time-varying data by listing data snapshots at multiple time points and overviewing the data in a timeline with automatic annotations of salient data features. Users can choose the desired techniques to select data snapshots and arrange them temporally, and score data attributes and points of interest to rank the annotations shown.

Utilizing external data sources, Contextifier [58] automatically generates annotated time series visualizations to contextualize stock news articles. It queries relevant news to the current article from the news corpus and time series stock data through APIs, identifies visually salient features in the data and notable events from relevant articles, and creates annotated stock visualizations integrating the textual and visual data. Meanwhile, Roslingifier [118] semi-automatically generates animated data presentations of time-series data using scatterplots based on a detailed analysis of 11

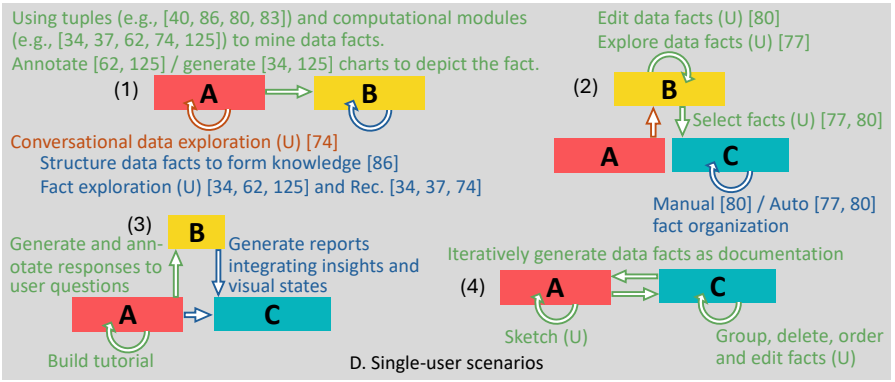


Fig. 12. Fact mining. (1) depicts papers that mine data facts and map facts to charts and textual descriptions for recommendation and exploration. CoInsight [77] and Notable [80] mine data facts to be selected and organized into data stories (2). Meanwhile, LEVA [154] uses LLMs to guide VDE and report generation (3). InkSight [83] enables user sketching to iteratively generate facts as documentation (4). (U) highlights user tasks in the processes.

data presentations. It detects significant data changes as events and presents them in a visually salient way using labeling, spotlighting, and tracing. Users can edit the event playback timeline and individual events of their labels and narratives.

Besides automatic extraction of data features to tell a story, researchers also created tools to support authoring stories (Figure 11(2)). Utilizing the links between data and texts, tools can suggest data attributes and highlight relevant data while users are typing and derive data and visualization based on the texts to assist analysis [16, 28]. For instance, by identifying data operations in texts via natural language processing, CrossData [28] calculates data derivation on the fly, such as mean values, to update existing data. With block-based story authoring, DataParticles [16] identifies data attributes and visual effects in the texts, such as "highlight top 5 rated drinks" to generate incremental visualizations based on the previous block.

Two domain-oriented papers feature the fluid transition from VDE to reporting to support discourse analysis [114] and make sense of geospatial phenomena, such as which factors influence regional voter preferences in elections [76]. They enable users to synthesize observation notes with relevant visualizations as PDF reports to facilitate result dissemination.

Three articles address the integration of VDE and reporting to support story iterations (Figure 11(3)). StoryFacets [99] and Gratzl et al. [48] utilize analysis trails to mediate data analysis and story generation. For instance, StoryFacets allows users to build a tree-structured trail, consisting of visualization cards and interactions connecting the cards, to capture meaningful analysis states; the states can be used to create dashboards and slideshows for different audiences, including data managers and clients. Analysis trails could be shared to support collaboration [48, 99].

Differently, utilizing the insight space (Figure 11(3)B), Chen et al. [24] proposed a framework enabling users to extract insights as structured representations of data (story slices) from the visual analytics system. Users can select, assemble, and arrange story slices in meaningful layouts to build the story for the general public.

4.3.8 Mining Data Facts for Visual Recommendation/Storytelling. There are two main methods to mine data facts, using tuples or computational modules (Figure 12(1)). QuickInsights [40] proposed a tuple structure, including insight type, score, subspace, breakdown, and measure, to uniformly represent and systematically mine various types of data facts. Within a filtered subspace, such

as filtering the sales data by country China, QuickInsights systematically breaks down data by a chosen dimension, such as a year, to extract facts, such as measuring sales trends and outliers in China throughout the years. The method has been adopted and extended in many later studies (e.g., [23, 117, 136]). Meanwhile, Tang et al. [129] proposed a score function capturing interesting insights by impact (importance of the subject) and significance (uncommonness). Building upon QuickInsights and Tang et al., MetaInsight [86] categorized data facts into commonness and exceptions to form structured knowledge and proposed a ranking algorithm to pick a set of MetaInsights with high individual scores and low redundancy.

In contrast, researchers also used computational modules [34, 37, 62, 74] to mine varied types of facts with specific ranking metrics, such as using the Pearson correlation coefficient to rank the linear relations between two variables [37]. For example, to recommend insightful charts to the client side, DataSite [34] built a server with scalable numbers of computational modules analyzing data from varied aspects and a scheduler scheduling the runtime of the modules based on estimated computational complexity and major data patterns.

To depict mined facts, researchers mapped facts to charts with embellishments, such as using scatter plots and regression lines to communicate correlational facts, and used templates to generate textual descriptions [34, 125], with some exceptions. For instance, LEVA [154] and SNIL [29] used **Large Language Models** (**LLMs**) to generate more natural descriptions of facts. These tools then recommend the depicted facts based on the data under exploration [34, 62, 125, 154], insights of interest [37], and conversational contexts [74]. For instance, Voder [125] generates data facts using heuristics and maps to corresponding charts and embellishments. Upon specifying a chart, users can explore its data facts with chart highlights upon hovering and view recommended facts with the corresponding charts under a different configuration of the selected data attributes, such as the sum and average of the attributes, as well as query for facts. To support non-technical users' VDE, Boomerang [74] recommends insights with charts through a conversational interface. Based on the conversational context, it receives insights from various modules and ranks them by interestingness, relevance, and timeliness for recommendation.

Furthermore, four research articles allow users to build data stories, connecting mined facts in linear [80, 83, 154] or hierarchical [77] structures. CoInsight [77] (Figure 12(2)) extracts three types of insights, point, shape, and compound insights, from **tables** and builds **an insight relation graph** based on their data scopes and the hierarchical table headers. Upon selecting insights of interest, users can explore the generated data story in a tree structure, preserving the insight relations. To ease the transition between VDE and data storytelling, Notable [80] (Figure 12(2)) incorporates data storytelling into **the computational notebook**. It mines six types of data facts from user-specified charts and uses templates to convey these facts through text descriptions and visual embellishments. Users can customize the facts and add facts of interest to the data story, which could then be automatically and manually ordered and exported as slides.

Further, LEVA [154] (Figure 12(3)) leverages LLMs to support VDE by generating tutorials based on view specifications, recommending insights during VDE based on data selections and specified tasks, and generating insight reports of user-selected analysis paths in LaTeX format. In contrast, InkSight [83] (Figure 12(4)) addresses the mixed-initiative generation of data facts by users **sketching** atop visualizations to specify data interest and InkSight suggesting data facts based on the sketch path and the selected data. Users can order, delete, group, and edit the facts to create a narrative.

4.3.9 Machine Learning for Data Fact/Story Generation. Researchers proposed machine learning techniques to support visualization insight and storytelling. Following QuickInsights [40], researchers extracted data facts from tabular [117, 136] and graph [23] data using tuples (a. orange

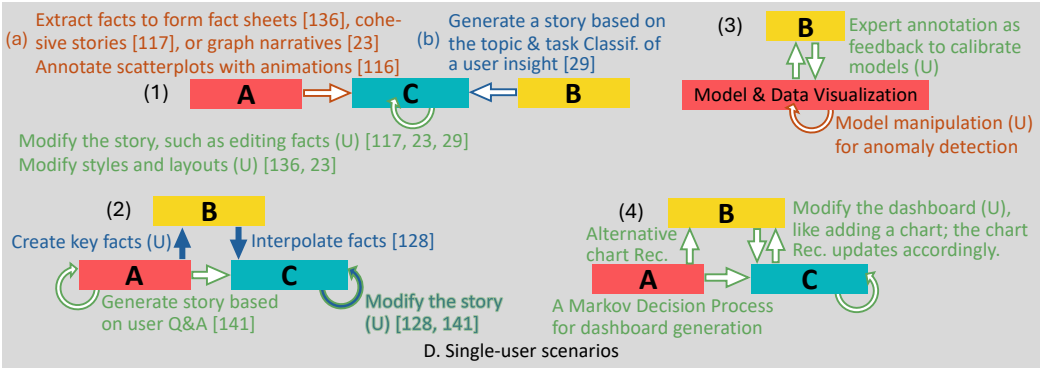


Fig. 13. Papers leveraging machine learning to build dashboards and data stories (Section 4.3.9). (U) highlights user tasks in the processes. (1a) Automated generation of fact sheets and data stories [23, 116, 117, 136]. (1b) Automated story generation from a user-provided insight [29]. (2) Automated data story generation guided by user input, such as user-created key facts [128] and user objectives elicited via Q&A [141]. (3) Expert annotation and model refinement to improve anomaly detection [1]. (4) Deep reinforcement learning approaches for generating insightful dashboards, using user adjustments as feedback to enhance dashboard quality [38, 78].

part of Figure 13(1)). To form **fact sheets**, DataShot [136] then extracts topics based on the fact subject and organizes the top-n facts of a topic with corresponding charts (using a decision tree model), descriptive texts (template), layout, and styling. Calliope [117] uses logic-oriented Monte Carlo tree search to connect data facts into **data stories** based on six types of coherent relations they have defined, such as elaboration and cause-effect. A story editor enables users to modify the storyline and edit individual facts.

Meanwhile, Calliope-Net [23] iteratively selects facts to build narratives based on the topic-evidence-explanation structure (maximizing fact score, narrative diversity, and network coverage), and generates visually pleasing, fact-annotated **graph narratives** by designing a layout algorithm. Users can choose the algorithm to regenerate the annotated graph, change styles and layouts, and modify graph encodings.

Similar to Calliope, Erato [128] and Socrates [141] utilize Monte Carlo tree search to **generate data stories**. However, they also enable users to steer the story generation process (Figure 13(2)). By letting users **create key facts** and converting facts to vectors via language processing, Erato [128] utilizes Monte Carlo tree search to interpolate transitional data facts to create a coherent story. Meanwhile, based on Pareto optimization, Socrates [141] picks **questions to elicit user preferences** and builds a Pareto frontier to generate stories that best satisfy all user objectives.

Differently, SNIL [29] generates **sports news stories** from users' **textual descriptions of an insight** (b. blue part of Figure 13(1)). Based on the topic and task classification of the insight, SNIL employs templates to narrate the story in episodes with appropriate charts depicting varied types of data facts and leverages LLMs to generate the episode title and description. Users can adjust the visual and textual contents of each episode and add, remove, and rearrange episodes. Focusing on model steering, Sintel [1] (Figure 13(3)) proposes a machine-learning framework incorporating human knowledge in **anomaly detection of time-series signal data**. It allows users to select anomaly detection pipelines, tune hyperparameters, and compare the methods with benchmarks. Users can also confirm, modify, and remove events as feedback to enhance model performance.

On the other hand, (deep) reinforcement learning is used to generate insightful resulting visualizations and prevails in mixed-initiative visualization generation. Utilizing Markov Decision

Processes, DashBot [38] and InsigHTable [78] propose deep reinforcement learning frameworks to build dashboards for flat and hierarchical tables, respectively (Figure 13(4)). DashBot uses chart diversity, parsimony, and insightfulness as rewards to make the resulting dashboard expressive and salient in data patterns. For InsigHTable, visualizations are embedded within the hierarchical table, with rewards to promote exploration and increase the quantity and diversity of insights. Users can edit charts within the dashboard or manually construct new charts to add to it; chart recommendations update accordingly to maximize rewards [38, 78].

Upon studying how analysts graphically annotate different types of data facts on scatterplots and the order in which they annotate these insights, Shi et al. [116] proposed to use the Markov decision process and reinforcement learning to automatically annotate scatterplots step by step in animation, which maximizes the number of statistical insights, rewards the logical order of annotations, such as from overview to the detail, and avoids visual clutter of annotations.

4.3.10 Summary and Discussion. Researchers proposed visualization designs to support multi-dimensional/actionable insights (Section 4.3.1), automations to extract data facts (Section 4.3.8), including tuple structures to systematically mine insights and computational modules to collect diverse types of insights, and techniques to support insight externalization (Section 4.3.2), such as annotating charts with texts and highlights and using pen and touch interactions for externalization. Nonetheless, chart errors can hinder people from drawing correct insights; multiple studies proposed effective methods to annotate chart errors and promote reflections on data patterns (Section 4.3.6).

Manual and automatic ways are devised to link text snippets and chart elements (Section 4.3.4), which can potentially benefit both the authoring and reading of visualization insight. To support sensemaking, presentation, and collaboration, insights, visualization/data, tags, comments, source materials, and other content are often linked in graph/tree structures, conveying analytic provenance (Section 4.3.5). To support insight exploration and analysis (Section 4.3.3), data-aware annotations link insights to the data or even propagate insights through the data hierarchy [41, 143], enabling users to retrieve insights from the visualization. Insights could also be retrieved using their metadata, such as timestamps, and assigned a lifetime and validity. Moreover, researchers studied various ways to recommend insights based on data attributes under exploration, interaction trails, selected insights of interest, and conversational context.

Insights are often organized as reports in a linear, logical manner to support communication and presentation (Section 4.3.7). Reports can take the form of overlaid layers (overlaying textual annotations over a base layer like a timeline), animations, and slideshows. Researchers also proposed techniques such as automatic suggestion of data attributes and visualization generation based on texts to ease the report-writing process. Many tools aim to integrate VDE and reporting, e.g., via analytic provenance, addressing iterative report writing with an easy transition between the two states. However, researchers have often focused on bottom-up analysis and reporting, while the top-down process, from reporting back to VDE and hypothesis-driven analysis, remains less explored.

Data mining and machine learning techniques are used on multi-dimensional data, including categorical, temporal, and numerical measures, to extract data facts and tell stories, except for Calliope-Net [23], which narrates graph data (Sections 4.3.8 and 4.3.9). Monte Carlo tree search is often used to connect data facts to form a coherent story. Human intervention, including creating keyframes, question answering, and providing textual descriptions of the insights, helps steer machines to generate data stories that satisfy user needs. Moreover, reinforcement learning is used to improve model performance based on user feedback to maximize action rewards and generate insightful dashboards. However, automated stories often lack the incorporation of domain

knowledge, hindering their applicability and audience engagement [128]. This limitation is also reflected in Table 2, where the papers in the last two columns largely fail to address insights into the domain.

Most techniques are developed for single users with desktop settings, with only a few supporting asynchronous collaboration. As data analysis in practice often involves multiple user roles [63], advancements in collaborative analytic scenarios are required. With the rise of extended reality technologies, opportunities also exist to explore visualization insight discovery in immersive analytical environments.

5 Modeling the Cycles of Visualization Insight

Figure 1 modeled the common flow of the generation, organization, and communication of visualization insights as an iterative and collaborative process involving diverse people and devices, summarized from the 92 reviewed papers. Reports/Presentations/Data stories (Figure 1C) are annotated visualizations that logically link insights for effective communication. Figure 1B highlights an insight repository to collect insights for organization, exploration, and recommendation. For instance, users can construct graphs connecting discovered concepts (with data/charts attached as provenance) for sensemaking [81, 151]. People and devices (Figure 1(D)) reflect the broad range of usage scenarios the process supports. In asynchronous collaborative analysis, an analyst may explore the data and organize insights to pass on to others for continued investigation [152]; researchers also designed tools to help storytellers craft data stories on desktop computers for audiences using mobile devices [117].

The transitions in Figure 1 can denote varied interactions across different scenarios, with examples shown in Figures 4–13. For instance, Transition f may indicate a story author revisits the insight repository to explore other story pieces [24], while for readers, it may signify the generation of insights from the story content. Transition c implies that (1) users are motivated by the insights to explore more data [24] or (2) users assign charts to depict varied types of insights [148].

Overall, this model can inform the design and evaluation of systems that support the discovery and communication of insights from data. It enables researchers and developers to articulate the processes their proposed methods support, justify the necessary transitions to support target users, and compare their methods with existing systems. For instance, Figure 11(3) contrasts systems that facilitate the fluid transitions between VDE and story creation. They address different usage scenarios, including single-user desktop settings [24] and asynchronous collaborative contexts [48, 99]. StoryFacets [99] and Gratzl et al. [48] utilize analysis provenance as a medium, allowing users to resume exploration and construct stories from meaningful historical states. In contrast, Chen et al. [24] proposed a dedicated space to store analytical findings, enabling their organization and annotation for story creation.

Additionally, this model has practical applications in educational settings [95]. Engaging students in hands-on VDE to discover insights, derive hypotheses, or validate them (Figure 1(a), (b), (c),and (i)) stimulates active learning and knowledge construction, fostering higher-level thinking skills aligned with Bloom's taxonomy [6], such as analysis and evaluation. Exchanging and exploring insights with peers (Figure 1(d)) encourages discussion and peer feedback. Communicating discoveries through reports (Figure 1(e), (f),and (g)) provides opportunities to assess students' critical-thinking and problem-solving skills.

5.1 Relation to Other Models

We compare our model with existing visual analysis frameworks related to sensemaking and knowledge building. Van Wijk's visualization model posits that users gain new knowledge by interacting with the visualization [138]. The increase in user knowledge depends on the visual

image, user perception and cognitive abilities, and their existing knowledge about the data. This clearly explains the VDE process and the role prior knowledge plays in knowledge generation. However, as seen from the reviewed papers, knowledge externalization and communication are equally important and should be recognized as an integral part of VDE.

Our model closely relates to the sense-making model by Pirolli and Card [100] (Figure 14(1)). User sensemaking is driven by both bottom-up and top-down analysis processes. In the bottom-up process, users collect information from data, extract and organize evidence, and build a hypothesis to present to the client, whereas during the top-down process, users re-evaluate the analysis by re-examining the story structure and searching for more evidence/information. In addition, connecting visualization and reporting (Figure 14(A) and (C)), our model indicates (1) the non-linear and iterative feature of analysis and (2) the possible authoring [16, 28] and automated [117, 136] process for data storytelling. Since the transitions in our model can take on varied meanings based on user roles and interaction modalities (Figure 14(D)) as mentioned earlier, our model is flexible to accommodate diverse analysis scenarios.

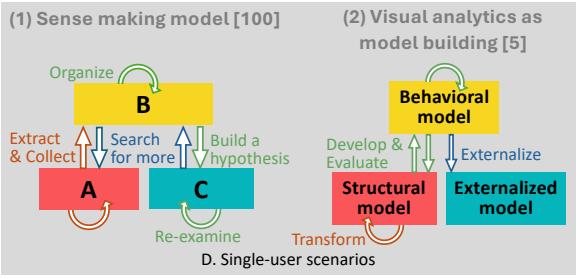


Fig. 14. How our model relates to frameworks by Pirolli and Card [100] and Andrienko et al. [5].

Figure 4(1) depicts how Sacha et al.'s knowledge generation model [107], featuring its exploration, verification, and knowledge generation loops relates to our framework. Similarly, to illuminate the human cognition processes involved in mixed-initiative human–computer collaboration, Green et al. [50] proposed a model encompassing the processes of information search & exploration, knowledge creation, and hypothesis generation & analysis as non-linear and interlinking. These two models focus more on the internal cognitive processes during visual analysis, while our model highlights the utilization of external artifacts to support knowledge building and hypothesis generation.

Andrienko et al. [5] consider visual analysis as model building, in which users gradually build their understanding of the subject of analysis (Figure 14(2)). Users iteratively develop and evaluate the model of the subject with the assistance of visualization, collecting provenance at the same time until the model is appropriate to be externalized for communication. The development of the subject model closely relates to the organization of insights collected from data; insights are not individualized but linked to reflect users' understanding and schematized to communicate their reasoning process.

Generally, closely relating to existing visual analysis and human cognition models, our model (1) stimulates the consideration of varied user roles and usage scenarios that could grant different meanings to the transitions between states, (2) highlights the role of external artifacts and domain information in discovering, organizing, and communicating visualization insights, and (3) accounts for computers' reasoning and hypothesis generation capabilities to automate insights into the domain and their reporting, enabled by the advancements in machine learning and LLMs.

6 Discussion and Future Research Directions

To study the research development in visualization insight, we reviewed 92 papers from the fields of visualization, HCI, and databases. Papers from visualization and HCI venues largely overlap in their emphasis on empirical studies and the design of interactive visualization systems. In contrast, papers published in database venues tend to differ in focus, more frequently emphasizing data mining techniques, such as automated insight extraction from data [40, 129, 142], and database

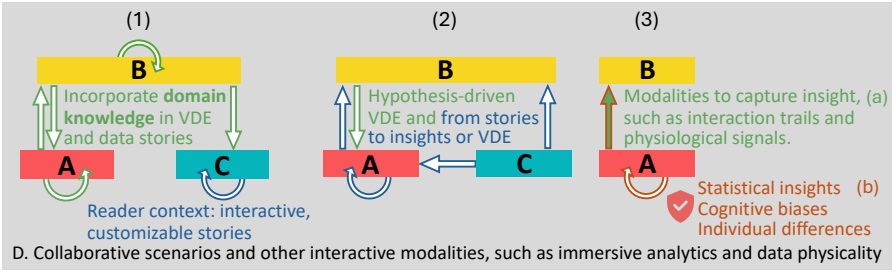


Fig. 15. An overview of future research directions.

management systems (e.g., annotation management [143]). Although interfaces are sometimes included for demonstration purposes (e.g., [37, 74]), these works generally place less emphasis on visual encoding choices or interaction design. This section expands on the future research directions identified in the literature review, which were briefly introduced in Sections 4.1.3, 4.2.6, and 4.3.10. Figure 15 summarizes these directions.

6.1 Advocating for Involving Domain Information in Visual Discovery

Theoretical papers highlighted the role contextual information plays in making visualization insights actionable and advancing our understanding of the domain [10, 64, 107]. Not just interpreting visualization insight within the domain but incorporating domain information in VDE is essential to facilitate **insights into the domain** (Figure 15(1)). For instance, prior work proposed design spaces to externalize personal knowledge about the data to trigger discussion and collaboration [82, 88].

Domain knowledge is structured, principle-based, and sometimes tacit [123]. McCurdy et al. [93] proposed a framework to elicit expert knowledge with a well-designed template to identify and contextualize implicit data errors. Close collaboration with domain experts regarding the utilization of their knowledge in VDE, like the study McCurdy et al. conducted, is crucial to devising visualizations that support insight into the domain.

Empirical studies on how the use of prior knowledge in a singular/collaborative setting could influence visual discovery can increase awareness of this important research topic. For instance, in a social context, Kim et al. [67] investigated how embedding others' expectations about the data in the visualization affects user recall and trust of the data, as well as their adjustment of beliefs about the data. Open access to research study data can significantly advance visualization insight research by enabling the community to conduct alternative analyses, compare findings across studies, build robust models with diverse datasets, and evaluate the effectiveness of proposed techniques in varied contexts.

6.1.1 Contexts in Data Storytelling. A decent amount of work has focused on automatically mining data facts and generating stories (Section 4.3.9). The proposed approaches can greatly ease the storytelling process and benefit information dissemination; users can interpret the news in their own way (Figure 1(f)) or perform follow-up searches for more personalized information (Figure 1(i) and (b)). However, bypassing contextual information or domain knowledge in story generation can limit the effectiveness of the data stories. For instance, Erato [128] revealed that automation could not create soulful stories that truly affect readers.

To tackle this, we can resort to automation and human input. The vast amount of information available on the Internet can enrich data with contextual details, situating data facts within a broader narrative to enhance their insightfulness and applicability (e.g. [17, 58]). Also, LLMs

can contextualize data facts drawing on their extensive knowledge database [154], though their limitations, like lack of common sense and hallucinations, need to be carefully considered. Meanwhile, humans contribute creativity, structured knowledge, and intuition, valuable in crafting in-depth, task-specific reports/stories. Nonetheless, effectively incorporating domain knowledge into automated storytelling remains an open research challenge.

Besides story authoring, from readers' viewpoints, automated stories often exhibit limited interactivity and lack engagement. Incorporating reader context into storytelling is another research direction [113]. For instance, other than a fixed, linear storyline, InsideInsights [92] introduces a hierarchical report structure, allowing reviewers to learn more details or abstracts as they wish.

6.2 Visualization Insight Discovery as a Top-Down Process

VDE is often driven by an explicit goal/hypothesis, though the resulting insights might not always be relevant to the initial goal (unexpected insights) [9, 107]. By studying hypothesis-based visual analysis, allowing users to express their expectations about the data before seeing them, Koi et al. [33] investigated what types of hypotheses users have, how these hypotheses affect data sensemaking, and so on. If not goal-driven, insights derived from VDE should be treated as hypotheses that require validation using a different dataset [64, 150]. Section 4.3.7 presented tools addressing the easy transition between VDE and reporting, but did not emphasize the top-down, hypothesis-driven aspect from reporting to VDE (Figure 15(2)). A relevant question is how to design the process to encourage users to collect evidence and validate insights against the domain. This could advance our understanding of the collaborative, iterative insight discovery process. For instance, LLMs could be used to retrieve supporting and opposing evidence from online sources to validate insights.

6.3 Insight Externalization beyond Note-Taking

Focusing on the textual representation of insight, most studies employed think-aloud protocols, digital pens/ keyboards, and templates to record insights. For instance, Kim et al. [68] found the advantage of a pen for quick thoughts and a keyboard for long chains of thoughts. In addition, the rapid advancement of LLMs in code generation, processing structured data and unstructured text, and producing structured outputs has made textual insights easier to extract and analyze. For example, research that emerged after the time span covered by this review explores the use of LLMs to link text and data [52], extract insights from data with visual evidence [32, 137], and organize insights by topics [137]. However, users may benefit from alternative interaction modalities to best express their insights, depending on individual learning styles (e.g., visual learners prevail at interpreting visuals while kinesthetic learners thrive through hands-on activities [133]) as well as the situation and task at hand [50] (a. green part of Figure 15(3)). For instance, Jianu and Laidlaw [61] allowed users to click protein nodes to construct a pathway to externalize their knowledge.

Meanwhile, in contrast to explicit knowledge, tacit knowledge is recognized as an intangible asset, which cannot be easily verbalized but is manifested in what experts do [112, 131]. For instance, a machine learning expert may intuitively adjust a data annotation schema to improve language model performance [18] or tune model parameters to optimize results efficiently. Within the context of VDE, tacit knowledge can manifest in user interaction trails and eye-tracking data. Andrienko et al. found that eye movements reveal tacit analytical activities, such as pairwise comparison and narrowing the focused region [4]. Examining the ways to record the externalization of tacit knowledge, such as logging meaningful VDE strategies, offers opportunities to benefit knowledge sharing within and beyond the domain.

Moreover, other objective physiological signals, such as pulmonary and electrodermal responses, can indicate moments of insight or frustration [19, 85]. Studies showed that spontaneous insights are often accompanied by bursts of activity in the right temporal lobe [19]. We can utilize these

natural signals to evaluate the effectiveness of visualizations in supporting insights and to cue moments of insight, for instance, to support social analysis.

6.4 Enhancing Cognitive Efficiency in Visual Discovery

Due to human cognitive limitations, visualization insight discovery requires machine support to mitigate biases and errors (b. orange part of Figure 15(3)), highlighting the need for adaptive and mixed-initiative analytic systems. *Promoting statistical insights.* Multiple studies in this review indicate users are not aware of or good with the statistical significance of visualization insights and often take spurious findings as true [31, 150], which also prevents data analysts from conducting open-ended VDE [2]. These point to the necessity of utilizing machines' capability with statistics to alleviate the multiple comparisons problem in VDE and increase the quality of insight. For instance, Savvides et al. [111] developed a statistical testing procedure for VDE as within-view and between-view problems to reduce false discoveries. Utilizing the Internet's knowledge base, machines can also identify additional datasets to validate insights [150].

Cognitive biases. Cognitive heuristics can speed up decision-making but can also lead to biases [135]. With cognitive limitations, humans inevitably bring their biases into VDE. For instance, the curse of knowledge enables domain experts to discover visual patterns that are puzzling for novices to uncover [144]. Dimara et al. [39] provided a taxonomy of cognitive biases in information visualization, revealing that most biases are not studied for their effects or relevance to visualization research. The impact of various cognitive biases on visual discovery is thus worth more research attention to investigate and alleviate visual discovery fallacies.

Individual differences. Individual differences can impact visualization insight discovery. Prior studies reported instances that personality traits, such as openness to experience [3] and locus of control [49] can affect the number of user insights. Thus, insight-based evaluation should account for these factors in experiment design [3]. Conversely, with a deeper understanding of these traits, visualization design can be adapted to enhance insight discovery [158].

6.5 Visualization Insight across Alternative Modalities and Collaborative Scenarios

A single-user desktop setting with a mouse and keyboard is the typical analytical environment of the surveyed papers. However, data analysis often involves people with different expertise using various tools [63, 87, 157]. In another sense, as Chinchor and Pike stated, all analyses are collaborative, even if the collaboration is with one's future self [30]. Typically, people work with different devices in different situations. Transferring analysis across devices with varied interaction modalities requires more research attention (Figure 15(D)), such as using voice to note ephemeral thoughts on charts via cutting-edge speech-to-text techniques. With multiple collaborators, tool design needs to consider empirical study findings (e.g., [63, 89]), such as separating private and public spaces and maintaining analysis provenance as an integral part of insights, allowing users to share intermediate analysis reports.

Immersive analytics. The development of extended reality has the potential to transform the way we think and interact with data. The 3D space around us affords the spatial organization of data items that users can leverage for data analysis tasks [59]. Annotation and information dissemination, as essential parts of data analysis, present design opportunities and challenges using novel interaction modalities, such as haptic, tangible, and tactile interactions [11, 42]. For instance, how to disseminate knowledge in public venues using various interaction paradigms is underexplored [11]. Moreover, the evaluation of these techniques is mostly limited to traditional metrics, i.e., task time and error, while overlooking the process of insight discoveries [11].

A search within venues focusing on novel interaction modalities, including the **International Symposium on Mixed and Augmented Reality** (**ISMAR**) and Conference on **Tangible,**

Embedded, and Embodied Interaction (TEI), using our search criteria did not return any results. This indicates opportunities for cross-disciplinary collaboration to address visualization insights using alternative modalities, instantiating, challenging, and expanding the proposed framework to capture diverse scenarios and draw a more comprehensive, empirically grounded picture of the cycles of visualization insight discovery.

Data physicality. Besides vision, being able to touch, taste, smell, and hear data opens more channels for users to experience insights [57]. With artistic intent, data physicalization can connect people on an emotional and soul level to trigger non-analytical insights, such as curiosity and awareness [8, 55]. However, little research has attempted to examine the effects of multi-sensory data representation in aiding insight generation [55]. Thus, we see great opportunities to design novel interaction techniques that use physical senses and spatial environments to explore their affordances in supporting data insights.

7 Conclusion

This systematic literature review analyzes and categorizes 92 papers relating to visualization insight. The reviewed papers span three areas of contribution (theoretical, empirical, and technological) and 16 research topics in total, leading to multiple takeaways: (1) Linking insights and relevant data/visual elements, such as interactive highlighting and annotation, can boost productivity in both reading and writing of insights. (2) Graph structures are frequently used to allow users to externalize their thoughts by connecting insights, tags, charts, data sources, and other content for sensemaking and communication. (3) Threats like cognitive biases and chart errors hinder the effectiveness of visual discovery, requiring more research endeavors. (4) Researchers often design tuple structures with scoring functions to mine and rank data facts and map facts to charts and highlights to convey insights. (5) Automation for data fact mining and story generation needs to incorporate domain information to enhance the results' impactfulness.

We summarized a model depicting the generation, exploration, and communication of visualization insights as a non-linear and iterative process. The model highlights the importance of incorporating domain information into the process and accommodates diverse user roles and usage scenarios. To facilitate the design and evaluation of visualization that aims to support visualization insight, the model can be used to justify the process a system supports and compare different systems. We then derived five exciting yet challenging research directions, including (1) incorporating domain information in VDE and report generation; (2) understanding the top-down process from insight/report to VDE to illuminate the iterative, collaborative process; (3) flexibility in enabling and capturing insight externalization beyond note-taking; (4) studying and addressing cognitive efficiencies in insight discovery; (5) designing insight for other human senses and utilizing alternative interaction modalities and the physical environment for visualization insight discovery.

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Appendices

A Initial Set of Representative Papers

Defining insight: [10, 19, 96]
Examining insight as an evaluation method: [51, 53, 96, 108, 109, 122]
Studying insight for visualization design: [31, 148]
Insight mining: [34, 40, 154]
Visualization insight related applications: [25, 34, 92, 151, 154]

B Topic Category Coding Criteria

Category	Definition
Theoretical research	
Defining/Categorizing Insights	Papers that address the definition of visualization insight or categorize data facts.
Formalizing the Insight Generation Process	Papers that conceptualize/standardize the insight generation process, such as how data transformations and user interactions lead to insights.
Empirical study	
Examining Insight as an Evaluation Method	Papers that propose a visualization evaluation method that involves examining insights.
Studying Insight for Tool Design	Papers that study how people use existing tools to generate and communicate insights to inform visualization design, not intending to improve the insight-based evaluation method.
How to Support Insight Externalization	Papers that study the mechanisms to support insight externalization.
Effects of Linking Texts and Charts	Papers that study how the visual link between texts and charts would affect users' preferences/task performance.
Insight Discovery Fallacies	Papers that study issues that could undermine insight discovery, such as human bias and misleading charts/texts.
Technological research	
Visual Representation to Support Insight Generation	Papers that focus on designing visualizations to support insight discovery.
Mechanisms to Support Insight Externalization	Papers that design methods to support users to externalize insights, such as through pen and touch.
Insight Management, Exploration, and/or Recommendation	Papers that propose insight recommendation methods or mechanisms to support user exploration of the collected insights.
(Automatic) Integration of Texts and Charts	Papers that propose methods to visually integrate text snippets and chart elements via, such as, coordinated highlighting and embedding texts within charts.
Linking Insights, Tags, Visualizations/Data, etc.	Papers that link at least three types of content, such as notes, charts, tags, concepts, and documents for sensemaking.
Supporting Reflections on Visualization/Insight	Papers that support annotating chart errors or the reflections on the quality of insights.
Report/Presentation Generation	Papers that focus on automatic or manual generation of data reports/presentations/stories.
Mining Data Facts for Visual Recommendation/Storytelling	Papers that propose methods to mine/extract data facts (for recommendation or story building).
Machine Learning for Data Fact/Story Generation	Papers that propose machine learning techniques to generate data facts or stories.

C Overview of Literature Coding Result

Here is a table summarizing the metadata and coding results of the 92 reviewed articles (Google sheet). In addition, an interactive MDS projection of the literature supports filtering and navigation to the papers' DOI pages (Tableau dashboard).

D Coding of Theoretical and Empirical Study Papers

	Insight level		Theory		Empirical study				
Articles	Insight into the data	Insight into the domain	Defining/Categorizing insights	Formalizing the insight generation process	Examining insight as an evaluation method	Studying insight for tool design	How to support insight externalization	Effects of linking texts and charts	Insight discovery fallacies
Chang et al. [19] (CG&A09)		✓	✓						
North [96] (CG&A06)		✓	✓		✓				
Law et al. [71] (VIS20)	✓		✓						
Chen et al. [27] (PacificVis09)	✓		✓						
Karer et al. [64] (TVCG21)		✓	✓						
Law et al. [72] (VIS20)		✓	✓			✓			
Smuc [121] (IV16)		✓	✓		✓				
Sacha et al. [107] (TVCG14)		✓		✓					
Vaidya and Dasgupta [134] (VIS20)		✓	✓	✓					
Gotz and Zhou [45] (IV09)		✓		✓					
Battle and Ottley [10] (TVCG24)		✓	✓	✓					
Saraiya et al. [108] (TVCG05)		✓			✓				
Saraiya et al. [109] (TVCG06)		✓			✓				
Gomez et al. [44] (VAST14)		✓			✓				
North et al. [97] (IV11)		✓			✓				
Smuc et al. [122] (CG&A09)		✓			✓				
Andrienko et al. [4] (CGF22)		✓			✓				
Guo et al. [51] (TVCG16)		✓			✓				
He et al. [53] (TVCG21)		✓			✓				
Kandogan et al. [63] (CG&A14)		✓				✓			
Choe et al. [31] (CG&A15)		✓				✓			
Yang et al. [148] (TOCHI14)	✓					✓			
Reda et al. [104] (CHI15)		✓				✓			
Kim et al. [68] (ISS19)		✓					✓	✓	
Mahyar et al. [89] (IV12)		✓					✓		
Goyal et al. [47] (CHI13)		✓					✓		
Zhi et al. [156] (CGF19)		✓						✓	
Shao et al. [115] (CHI24)	✓							✓	
Lee-Robbins et al. [75] (TVCG22)		✓						✓	
Zgraggen et al. [150] (CHI18)	✓								✓
Sarma et al. [110] (CHI24)	✓								✓
Xiong et al. [144] (TVCG20)	✓							✓	✓
Stokes et al. [126] (TVCG24)	✓	✓						✓	✓
Zheng and Ma [155] (PacificVis22)		✓						✓	✓

[a]ACM conference on interactive surfaces and spaces.

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